

Diagnostic performance of quantitative measures from FDG-PET/CT, FEC-PET/CT and DW-MRI in the detection of lymph node metastases in endometrial and cervical cancer.

Authors: Ben King², Tara Barwick², Nishat Bharwani¹, Andrea Rockall^{2*}

Affiliations: 1. Department of Imaging, Imperial College Healthcare NHS Trust. 2. Department of Cancer and Surgery, Imperial College.

Abstract: *Background:* Prognosis for cervical and endometrial cancer patients with lymph node metastases is poor. Accurate staging of lymph nodes is essential to determine an appropriate treatment plan. Surgical lymphadenectomy is considered the most accurate technique for lymph node staging but is controversial as it has not been proven to be of therapeutic benefit and is associated with morbidity. Imaging techniques that are sensitive and specific in the detection of nodal metastases are highly desirable, but none have yet exhibited sufficient diagnostic performance to warrant replacement of lymphadenectomy in routine practice. Few past studies have discussed the utility of quantitative measures from pre-operative scans and have presented heterogeneous results. The present study investigates the diagnostic performance of quantitative measures derived from FDG-PET/CT, FEC-PET/CT and DW-MRI scans in the detection of lymph node metastases in endometrial and cervical cancer patients.

Materials & Methods: One hundred and sixty-two endometrial and cervical cancer patients were recruited to this multicentre prospective clinical trial. FDG-PET/CT, FEC-PET/CT and DW-MRI scans were obtained and SUVmax on PET/CT and ADCmean on DW-MRI of suspected lymph nodes in the pelvic and para-aortic regions were quantified. Histology by lymphadenectomy was considered the reference standard. ROC analysis was undertaken to estimate diagnostic performance of each imaging modality. Interobserver agreement of the quantitative parameters was assessed.

Results: SUVmax, nodal-to-tumour ratio (NTR) and SUVmax-to-ADCmean ratio (STAR) was significantly higher in malignant lymph nodes than benign lymph nodes in the pelvis on FDG-PET/CT in endometrial cancer. No significant difference in SUVmax or ADCmean, or NTR of either of these measures, was observed on FEC-PET/CT or DW-MRI, respectively, in endometrial cancer. STAR was significantly elevated in malignant lymph nodes of the pelvis on FEC-PET/CT in endometrial cancer. No significant difference was observed between any measure in cervical cancer although this may be due to low numbers of nodes available for analysis and generally earlier stage. In endometrial cancer, NTR of SUVmax on FDG-PET/CT exhibited the highest specificity but a low sensitivity (94% and 65% respectively). SUVmax, NTR and STAR on FEC-PET/CT produced moderate sensitivities and specificities. DW-MRI performed poorly with ADCmean and NTR producing low sensitivity and moderate specificities. Low numbers of analysed nodes limited analysis in the cervical cancer cohort but NTR and STAR on FDG-PET/CT produced the best diagnostic performance of the three imaging modalities. Interobserver agreement was excellent on FDG-PET/CT and FEC-PET/CT but poor on DW-MRI.

Conclusions: The diagnostic performance of quantitative measures from FDG-PET/CT, FEC-PET/CT and DW-MRI alone are not sufficient for accurate lymph node staging in endometrial and cervical cancer. Specific quantitative measures from FDG-PET/CT, FEC-PET/CT and DW-MRI may be beneficial in aiding the differentiation of metastatic lymph nodes in endometrial and cervical cancer.

Background

The prognosis for both cervical and endometrial cancers is associated with presence of lymph node metastasis. In endometrial cancer the 5-year disease free survival is estimated at 90% for patients without lymph node metastasis, 60-70% for those with malignant pelvic lymph nodes and 30-40% with malignant para-aortic lymph nodes (1). In cervical cancer the 5-year overall survival is estimated at 92% with localised disease and 58% with lymph nodes metastasis (2). Treatment options differ for patients with metastatic lymph nodes. Patients with localised disease will receive surgery as their primary treatment whereas patients with malignant lymph nodes will typically be referred for chemoradiotherapy.

Accurate staging of lymph nodes is therefore critical in treatment planning and improving prognosis for patients. Although surgical lymphadenectomy is considered the

most accurate technique for the detection of lymph node metastases, systematic lymphadenectomy is controversial as the procedure carries significant risks of lymphoedema, lymphocyst formation, and injury to surrounding organs and tissues (3, 4). Furthermore, this procedure has not been shown to reliably convey any significant survival benefit and is therefore not widely recommended as a therapeutic option (5). Accurate pre-operative staging of lymph nodes would therefore have a significant impact on reducing morbidity of unnecessary lymphadenectomy procedures, surgical time, and cost of staging.

Multi parametric magnetic resonance imaging (MRI) is currently considered the optimal imaging technique for local tumour staging of endometrial cancer through assessment of myometrial and cervical invasion but the major limitation of all imaging techniques is poor detection of metastatic lymph nodes (1). Morphological signs of nodal involvement, including size criteria, on CT and MRI

vary and are unreliable markers of malignancy since with size criteria there is a trade-off between sensitivity and specificity (6). Tumour may have spread to non-enlarged nodes (micrometastases) and nodes can enlarge due to benign conditions. Functional imaging techniques that can detect changes in the microenvironment and metabolic properties of tissues pose an alternative strategy for diagnosis of lymph node status.

Integrated positron emission tomography/computed tomography (PET/CT) with ^{18}F -fluoro-2-deoxy-D-glucose (^{18}F -FDG), a glucose analogue, allows combined anatomical and functional imaging. Visual diagnosis of nodal status in endometrial and cervical using FDG-PET has been evaluated in a number of small, mainly retrospective, clinical trials. Reported sensitivities of FDG-PET/CT for nodal involvement are heterogeneous (63%-80%) whilst specificity and accuracy are high (93%-97% and 94% respectively) (5, 7). Low sensitivity of FDG-PET/CT is considered to be the main limitation of the imaging modality due to the limited spatial resolution of PET imaging (8).

Maximum standardised uptake value (SUVmax) is a semi-quantitative measure derived from FDG-PET/CT scans that represents relative glucose uptake in tissues. SUVmax of the primary tumour has been shown to be higher in patients with higher grade endometrial cancer, myometrial invasion, cervical invasion, and lymph node metastasis (9, 10). Most studies to date have used visual assessment of the level of uptake in nodes compared to background and to date reports on the utility of SUVmax in the detection of nodal involvement are limited. Kim *et al.* retrospectively assessed SUVmax in histologically confirmed benign and malignant lymph nodes and reported moderate sensitivity (69%) and high specificity (92%) when using an SUVmax cut-off of 2.8 (11).

Diffusion-weighted MRI (DW-MRI) detects changes in the diffusion properties of water in tissues which is primarily influenced by cellular density and the nuclear-to-cytoplasm ratio and can therefore enhance subtle underlying abnormalities in tissues. DW-MRI is established for routine use in endometrial cancer staging and is shown to be promising in diagnosing extent of myometrial invasion adjunct to conventional T2-weighted imaging (12-14). There has been some research investigating the utility of the quantitative DW-MRI measure apparent diffusion coefficient (ADC) in the diagnosis of lymph node metastasis. The measurement of ADC from DW-MRI allows for quantitative assessment of cellular density. Whilst some studies report metastatic nodes have a lower ADCmean than benign nodes other studies have not suggesting there is a significant overlap in ADC values between benign and malignant nodes with no reliable cut-off (15-17). Furthermore, in endometrial cancer, retrospective studies have also produced heterogeneous results with some showing ADC is significantly different in metastatic and benign lymph nodes and others finding no difference (15, 18).

Choline is a critical component of cell membranes. In rapidly dividing endometrial cancer cell lines increased uptake of choline has been observed (19). Radiolabelled

choline (^{11}C or ^{18}F choline) can be incorporated with PET for functional imaging of cell membrane metabolism in vivo and has established use in detection of malignant lymph nodes in prostate cancer exhibiting a high specificity and moderate sensitivity (20). In small feasibility studies, ^{11}C -choline PET has shown similar diagnostic performance to FDG-PET/CT in the identification of gynaecological tumours (21). Additionally, ^{11}C -choline PET/CT had high accuracy for nodal staging in uterine carcinoma (22). Despite positive early findings no large prospective studies have reported the performance of ^{18}F -fluoroethyl-choline-PET/CT (FEC-PET/CT) in the diagnosis of nodal involvement in endometrial or cervical cancer. Furthermore, no investigations into the diagnostic utility of nodal SUVmax on FEC-PET/CT in gynaecological cancers have been conducted to date.

The present study aims to investigate the ability of quantitative measurements from FDG-PET/CT, FEC-PET/CT and DW-MRI in differentiation of metastatic and benign lymph nodes and determine the diagnostic utility of these imaging techniques.

This study forms part of the secondary analysis of the prospective UK MAPPING study.

Materials & Methods

Patients

This study analyses data from the MAPPING study – a multicentre, prospective study investigating the diagnostic accuracy of FDG-PET/CT, FEC-PET/CT and DW-MRI in the detection of nodal metastases. Ethics approval was obtained (Research Ethics Committee reference number 11/LO/1465) and all participants gave written informed consent.

Participants were recruited from 5 gynae-oncology tertiary referral centres in National Health Service (NHS) hospitals in England. Eligible patients were over the age of 18 years and had newly diagnosed histologically confirmed cervix or endometrial cancer. Pre-treatment FIGO (International Federation of Gynaecology and Obstetrics) stage was established on the basis of clinical examination and standard of care imaging (CT and MRI). Patients with cervix cancer were eligible if the pre-treatment FIGO stage was stage 1B1 or higher if still considered operable by the multidisciplinary team. Patients with endometrial cancer were eligible if histological grade was 3 with myometrial invasion on MRI or histological grade was 1 or 2, with suspected FIGO stage II or above on MRI. Patients were ineligible if they were unable to provide written informed consent, were pregnant or had contra-indications to MRI or PET/CT.

Participants were identified at the multidisciplinary team meetings and outpatient clinics, by research teams at each site who took informed consent from consecutive unselected eligible patients for the study DW-MRI and FDG-PET/CT. Separate consent was obtained for FEC-PET/CT. Age, tumour grade, pre-and post-treatment FIGO stage and the final treatment received were collected.

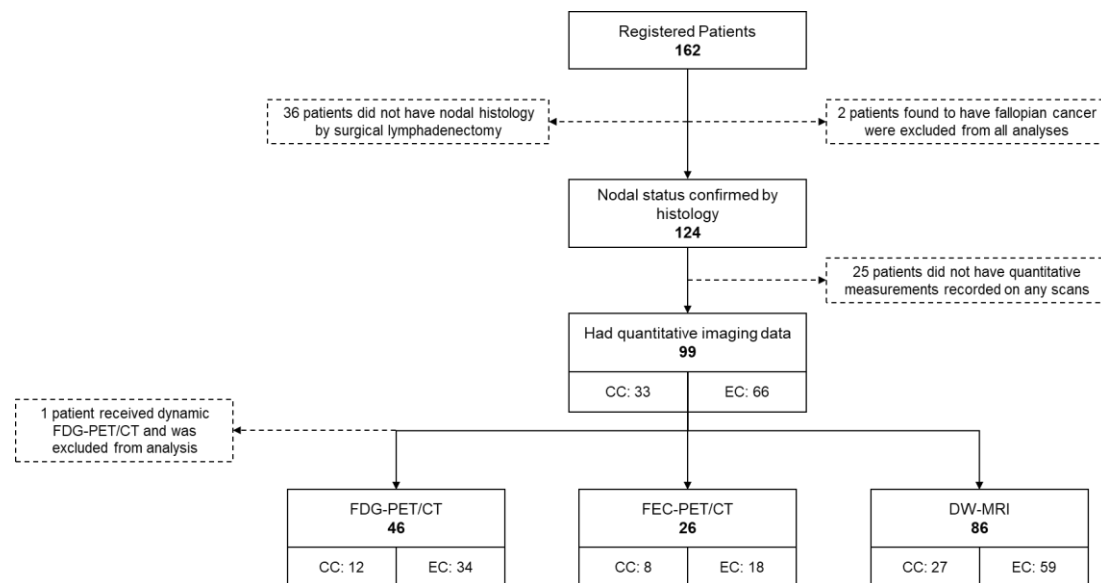


Table 1 – CONSORT trial flow diagram of this secondary analysis of the MAPPING study. Patients were eligible for analysis if they had histology of the pelvic or para-aortic lymph nodes by surgical lymphadenectomy and quantitative imaging data from FDG-PET/CT, FEC-PET/CT or DW-MRI. CC, cervical cancer; EC, endometrial cancer.

One-hundred and sixty-two patients were enrolled in the study. Of these, thirty-three cervical cancer and 66 endometrial cancer patients met the primary reference standard of nodal histology, had quantitative imaging data, and were therefore eligible for analysis (Fig. 1).

¹⁸F-FDG PET/CT protocol & ¹⁸F-FEC PET/CT protocol

FDG and FEC PET/CT scans were acquired using a standardised protocol based on UK NCRI PET Research Network guidance at accredited centres. Participants fasted for at least 4 hours prior to FDG injection (median 370 MBq, range 217-436 MBq) and scanning only performed if blood glucose was < 10 mmol/L. Fasting was not required prior to FEC injection (median 293 MBq, range 209-369 MBq). Both tracers were classified as investigational medical products. FDG scans were acquired at a median of 60 mins post injection (range 57-80) and FEC PET/CT scans at a median 60 mins (range 55-74) on separate days. Low dose CT was performed followed by emission study (base of skull to upper thighs) as 3D acquisitions. Images were reconstructed using ordered subset expectation maximisation and the PET images were attenuation corrected using the CT data.

DW-MRI protocol

Standard of care conventional MRI scan for staging was performed as per local protocol which included a minimal dataset of MR sequences. Depending on the timing of patient recruitment, the study specific DW-MRI was performed either at the time of standard of care MRI or at a second attendance. The MRI field of view included the pelvis and para-aortic regions up to the level of the left renal vein. For nodal evaluation, readers had a minimal dataset that included axial T1, axial T2 and axial diffusion weighted MRI (b values 0, 300, 600, 900 and 1200) with associated calculated ADC maps.

Surgical staging and pathologic evaluation

The primary reference standard was confirmed nodal histology by surgical lymphadenectomy. The surgical lymphadenectomy was undertaken as per the MAPPING study protocol (EudraCT 2011-001290-78) and anatomical location, presence of metastatic disease, and lymph node dimensions in each region (right pelvis, left pelvis, and para-aortic) were recorded. A region was determined to be positive for nodal metastases with the presence of at least one malignant node at histology.

Image evaluation

All images were read by two central MRI and PET readers; the central reads were co-ordinated through the UK NCRI PET core lab and the trials unit. All readers were accredited radiologists that were core members of the gynaecology multi-disciplinary team and/or PET/CT experts. Readers were made aware of the clinical diagnosis (endometrial or cervical cancer) from the pre-operative biopsy results, as per standard clinical practice. Readers were masked to other imaging studies and the final nodal histology.

FDG-PET/CT & FEC-PET/CT image evaluation

The anatomical location of any focally increased tracer uptake that was higher than background adjacent tissue corresponding to a node of any size on the CT was recorded. An elliptical ROI was manually placed on the slice with the highest activity and a measurement of SUVmax was recorded. SUVmax of the primary tumour was also measured by a single central expert PET reader by placing a regular ROI over the primary tumour on the slice with the highest activity on a dedicated PET/CT workstation (Hermes Medical Solutions).

DW-MRI image evaluation

Readers were aware of the nodal size from the standard of care MRI, and DW-MRI was additive to the standard conventional imaging. Nodes were identified on the high b-value ($b = 1200 \text{ mm}^2/\text{s}$) DW-MRI as non-continuous high signal intensity (SI) round or ovoid structures that corresponded with a node on the anatomical images. Short-axis diameter on MRI was measured for each identifiable node. The ADCmean of identified nodes were measured by manually placing a regular ROI over nodes greater than 5 mm short axis diameter, avoiding the margin and noted with anatomical location. ADCmean of the primary tumour, if visible, was also measured in each patient by both readers in an identical fashion.

Data analysis and manipulation

The highest SUVmax, for FDG-PET/CT and FEC-PET/CT, and lowest ADCmean, for DW-MRI, from the left pelvis, right pelvis, and para-aortic regions from each reader was selected for analysis. In cases where both readers provided measurements in a region, the readings were averaged and included in the final analysis. Where only one reader provided a measurement in a region, this measurement was included in the final analysis. The mean ADCmean of the primary tumour from both readers was calculated and included in the final analysis. Reads in regions that had corresponding nodal histology were included in the analysis.

Statistical analysis

Statistical analysis was performed in R. Patients were stratified into two cohorts by their diagnosis of endometrial or cervical cancer. Nodal-to-tumour ratios (NTR) were calculated by dividing the nodal quantitative measure (SUVmax or ADCmean) by the quantitative measure from the primary tumour. SUVmax-to-ADCmean ratio (STAR) was also calculated by dividing the SUVmax by the ADCmean of the corresponding node. Additionally, readings from left pelvis and right pelvis were combined into a 'pelvic' group and readings from left pelvis, right pelvis, and para-aortic regions were combined into an 'all' group for analysis. A Mann-Whitney U-test was performed to compare the mean SUVmax, mean ADCmean, nodal-to-tumour ratios, and SUVmax-to-ADC ratios as measured in histologically confirmed benign versus malignant lymph nodes in each region and group. P-values were adjusted using the false-discovery rate (FDR) method to produce adjusted q-values.

A generalised linear regression model was employed to determine if SUVmax or ADCmean was correlated with SA diameter for each measure for each imaging modality. Logistic regression was used to determine if SUVmax and ADCmean were correlated with histology of the lymph nodes. When correlation was found between SA diameter and the quantitative measure, the model was adjusted for short-axis diameter of the corresponding lymph node. Diagnostic odds ratios (DOR) were calculated.

To determine diagnostic performance for the detection of lymph node metastasis of these measures receiver-operating characteristic (ROC) analysis was undertaken using the package 'cutpointr' and 'pROC'. Optimal cut-off

Characteristic	n = 99
Median age (range), years	63.5 (24-80)
Median BMI (range), kg/m ²	28 (18-50)
Histology	
Cervical (n = 33)	
Adenocarcinoma	11 (33%)
Adenosquamous	6 (18%)
Squamous cell	15 (45%)
Unknown	1 (3%)
Endometrial (n = 66)	
Endometrioid	35 (53%)
Serous/clear cell	26 (39%)
Other	2 (3%)
Mixed (2 or more)	3 (5%)
Surgical Procedure	
Laparoscopic	42 (42%)
Robotic	31 (31%)
Open	26 (26%)
Lymphadenectomy	
Left pelvic	67 (68%)
Right pelvic	73 (74%)
Para-aortic	26 (26%)
FIGO stage (on MRI)	
Cervical (n = 33)	
IB1	29 (88%)
IB2	1 (3%)
IIA1	2 (6%)
IIB	1 (3%)
Endometrial (n = 66)	
IA	18 (27%)
IB	16 (24%)
II	9 (14%)
IIIA	3 (5%)
IIIC	14 (21%)
IVB	5 (8%)
Unknown	1 (2%)

Table 1 – Demographics of patients enrolled onto the MAPPING study and eligible for analysis by meeting the primary reference standard of surgically confirmed nodal histology.

values were calculated for each measure for each imaging modality and predicted nodal status classified by these cut-offs. Sensitivity, specificity, area under the curve (AUC), positive predictive values (PPV), negative predictive values (NPV), positive likelihood ratio (PLR), negative likelihood ratio (NLR), Youden's J index, and accuracy were calculated.

Interobserver agreement for each imaging modality was evaluated by calculating intraclass correlation coefficients (ICC) for endometrial and cervical cancer cohorts

separately using the package 'irr'. A two-way model was used and agreement between readers was evaluated. Confidence level of the interval was set at 95% and confidence intervals calculated.

For all analyses, p-values or q-values of less than 0.05 were considered statistically significant.

Results

Demographics

The demographics of patients eligible for analysis are summarised in Table 1. A total of 99 patients were eligible for analysis: 33 cervical cancer patients and 66 endometrial cancer patients. 67 and 73 patients received left pelvic and right pelvic lymphadenectomy, respectively, while just 26 patients underwent para-aortic lymphadenectomy. 88% of cervical cancer patients were diagnosed as FIGO stage IB1 and no patients in this cohort had a FIGO stage greater than II. In the endometrial cancer cohort, FIGO stage ranged from IA to IVB (Table 1).

SUVmax and ADCmean in metastatic and benign lymph nodes

The SUVmax from FDG-PET/CT in metastatic and benign lymph nodes was assessed in both endometrial and cervical cancer. In endometrial cancer, SUVmax was significantly higher in malignant lymph nodes than in benign lymph nodes in the pelvis (Fig. 2A2). When all lymph nodes were grouped together a significant difference in SUVmax was still observed (Fig. 2A1). The mean SUVmax from benign lymph nodes in all regions was 3.7 compared to 9.8 in malignant lymph nodes ($q < 0.01$). Excluding para-aortic lymph nodes, the mean SUVmax of malignant lymph nodes in the pelvis was 10.7 compared to 3.4 in benign lymph nodes ($q < 0.01$). The SUVmax in malignant para-aortic lymph nodes was not significantly higher than in benign nodes.

Although the mean FEC-PET/CT SUVmax in malignant nodes tended to be higher than benign nodes this did not reach statistical significance in any region(s) (Fig. 2B). The difference in SUVmax between malignant and benign lymph nodes in the pelvis (Fig. 2B2) came close to significance ($p = 0.08$) after multiple hypothesis testing. From DW-MRI, there was no significant difference in ADCmean between benign and metastatic lymph nodes in any region(s) (Fig. 2C).

No patients had para-aortic lymph node histology in the cervical cancer cohort; analysis was restricted to the pelvic region. There was no significant difference in the quantitative measures from FDG-PET/CT, FEC-PET/CT or DW-MRI between benign and metastatic lymph nodes in the pelvis (Fig. S1).

Nodal-to-tumour ratio

The difference in nodal-to-tumour ratio (NTR) of SUVmax or ADCmean between metastatic and benign lymph nodes was evaluated for each imaging modality. The NTR of SUVmax from FDG-PET/CT was significantly higher in malignant than benign nodes in the pelvis (Fig. 3A2).

Mean NTR in benign lymph nodes in the pelvis was 0.24 whilst in metastatic lymph nodes the mean NTR was 0.50. Mean NTR was higher in metastatic lymph nodes in all regions but the difference was not significant in any other than the pelvis. On FEC-PET/CT, mean NTR was higher in malignant lymph nodes in the pelvis although the difference was not significant (Fig. 3B). On DW-MRI, no difference was observed in NTR between malignant and benign lymph nodes in any region (Fig. 3C).

In the cervical cancer cohort, there was no significant difference in NTR from FDG-PET/CT, FEC-PET/CT or DW-MRI between benign and metastatic lymph nodes in the pelvis (Fig. S2).

SUVmax-to-ADC ratio

SUVmax-to-ADC ratio (STAR) was calculated for both FDG-PET/CT and FEC-PET/CT using ADCmean from DW-MRI and compared in metastatic and benign lymph nodes. On FDG-PET/CT, STAR was significantly higher in metastatic lymph nodes of the pelvis but not of the para-aortic lymph nodes in endometrial cancer (Fig. 4A). The mean STAR in benign lymph nodes in the pelvis was 4.1×10^{-3} whilst in metastatic nodes the mean STAR was 14.5×10^{-3} . Similar results were found in FEC-PET/CT. In the pelvis, STAR was found to be significantly higher in malignant nodes than in benign nodes with a mean STAR of 6.1×10^{-3} and 2.7×10^{-3} respectively (Fig. 4B2).

There was no significant difference in STAR from either FDG-PET/CT or FEC-PET/CT combined with DW-MRI between benign and metastatic lymph nodes in the pelvis in the cervical cancer cohort (Fig. S3).

Diagnostic performance of FDG-PET/CT

The diagnostic performance of FDG-PET/CT SUVmax, NTR and STAR in the detection of metastatic nodes cancer were evaluated using ROC analysis. The results for the endometrial cancer cohort are summarised in Table 2 and the corresponding ROC curves are shown in Figure 5A. NTR showed particularly good diagnostic performance with high accuracy, PPV and PLR albeit with low sensitivity. Linear regression analysis showed SUVmax was significantly correlated with short-axis (SA) diameter on FDG-PET/CT. SUVmax was significantly correlated with nodal histology with a diagnostic odds ratio (DOR) of 1.38 ($p < 0.01$). Furthermore, correlation between SUVmax and histology was maintained after adjustment of the model for SA diameter and produced a DOR of 1.38 ($p < 0.05$).

In cervical cancer, low numbers of analysed nodes limited analysis. ROC analysis was undertaken in the FDG-PET/CT cohort. The results are summarised in Table S1 and corresponding ROC curves plotted in Figure S4B. STAR performed particularly well and produced the greatest Youden's J index and exhibited the highest accuracy.

Diagnostic performance of FEC-PET/CT

Results from the ROC analysis of the FEC-PET/CT cohort in endometrial cancer are presented in Table 2 and corresponding ROC curves in Figure 5B. Stratifying patients against the SUVmax cut-off of 2.7 produced the highest

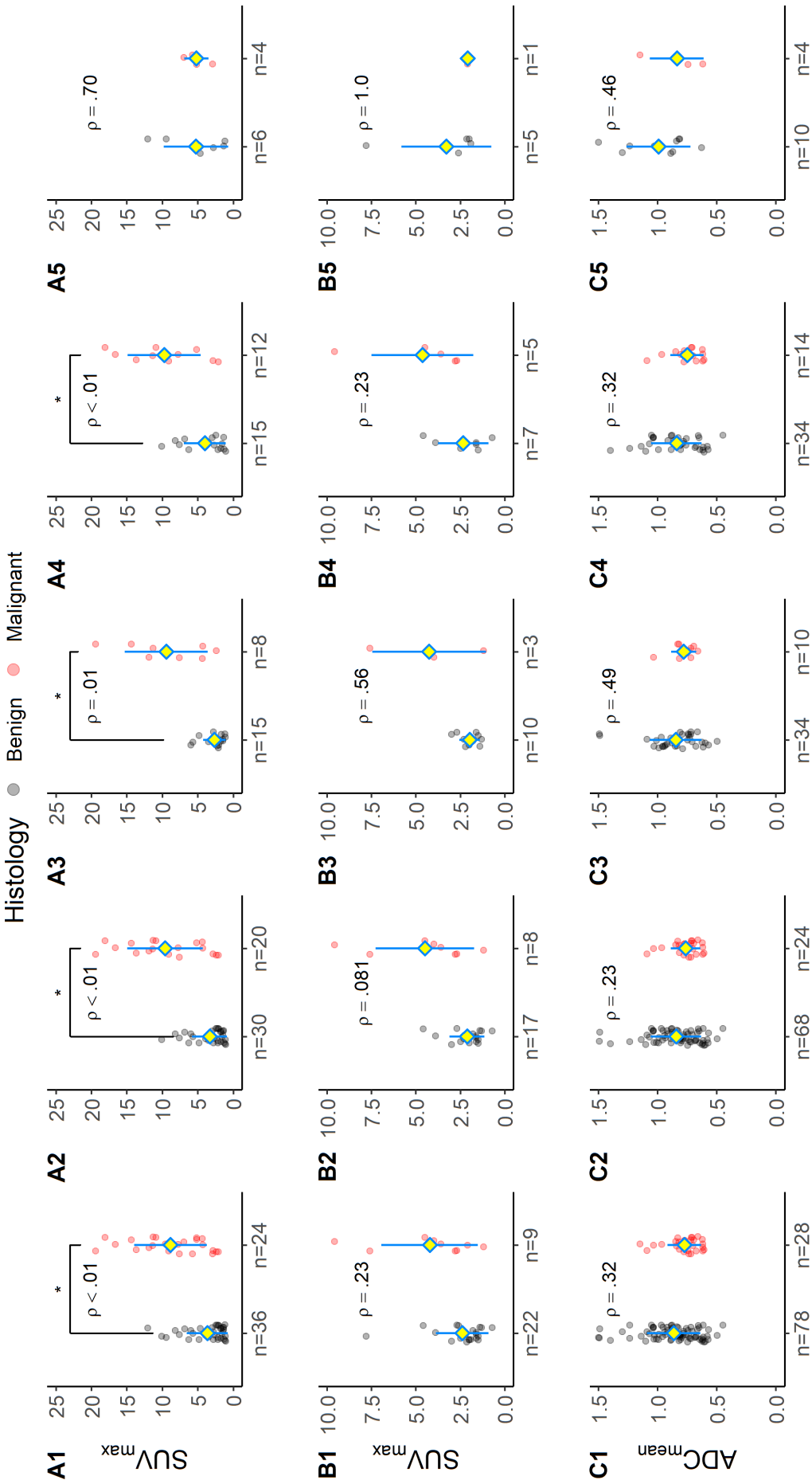


Figure 2 - Differences in SUVmax on (A) FDG-PET/CT and (B) FEC-PET/CT and in ADCmean (mm²/s) on (C) DW-MRI in benign and malignant lymph nodes in patients with endometrial cancer. Column 1 represents all nodes pooled together and column 2 represents the pelvic nodes pooled together. Column 3 represents left pelvic nodes, column 4 right pelvic nodes, and column 5 represents para-aortic lymph nodes. Yellow points represent the group mean and the blue lines represent ± 1 standard deviation. Significant difference between group means, calculated by Mann-Whitney U-test, is represented with a bracket and asterisk (*; $p < 0.05$). FDR adjusted p-values are given as p-values. Numbers of nodes analysed in each of the benign and malignant groups are shown on the x-axis.

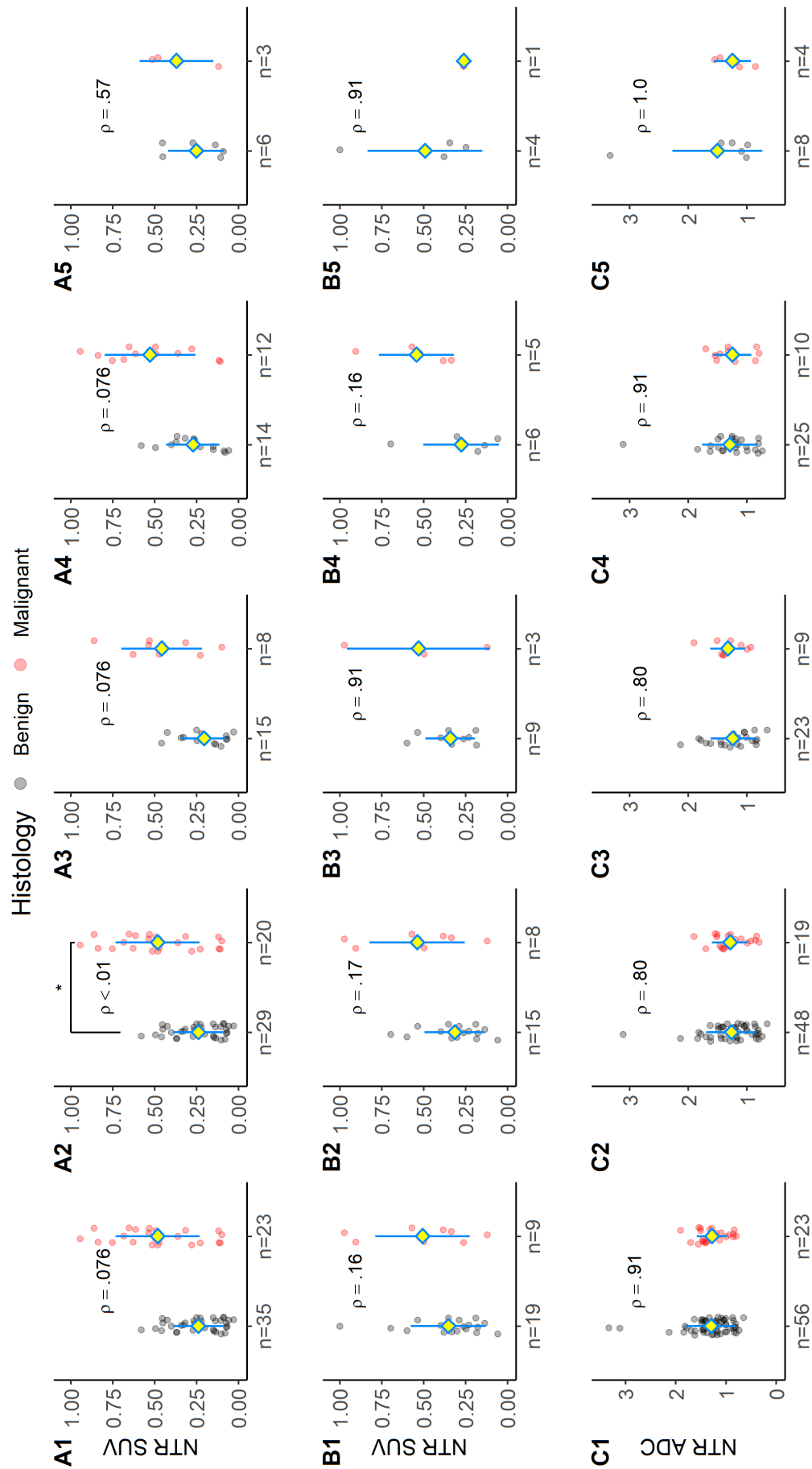


Figure 3 - Nodal-to-tumour ratio (NTR) of SUVmax on (A) FDG-PET/CT and (B) FEC-PET/CT and of ADCmean (mm²/s) on (C) DW-MRI in benign and malignant lymph nodes in patients with endometrial cancer. Column 1 represents all nodes pooled together and column 2 represents the pelvic nodes pooled together. Column 3 represents left pelvic nodes, column 4 right pelvic nodes, and column 5 represents para-aortic lymph nodes. Yellow points represent the group mean and the blue lines represent ± 1 standard deviation. Significant difference between group means, calculated by Mann-Whitney U-test, is represented with a bracket and asterisk (*, $p < 0.05$). FDR adjusted p-values are given as p-values. Numbers of nodes analysed in each of the benign and malignant groups are shown on the x-axis.

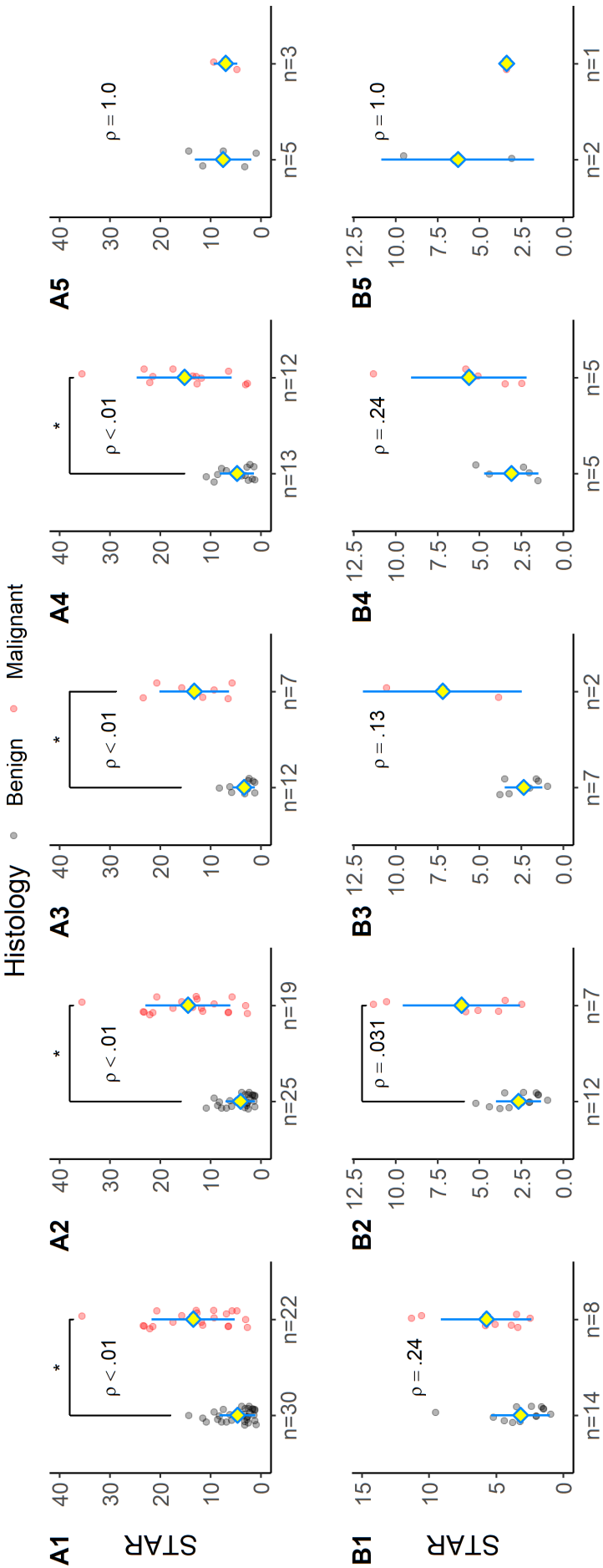


Figure 4 - SUVmax-to-ADC ratio (STAR) from combined (A) FDG-PET/CT and DW-MRI and (B) FEC-PET/CT and DW-MRI in benign and malignant lymph nodes in patients with endometrial cancer. Column 1 represents all nodes pooled together and column 2 represents the pelvic nodes pooled together. Column 3 represents left pelvic nodes, column 4 right pelvic nodes, and column 5 represents para-aortic lymph nodes. Yellow points represent the group mean and the blue lines represent ± 1 standard deviation. Significant difference between group means, calculated by Mann-Whitney U-test, is represented with a bracket and asterisk (*; $p < 0.05$). FDR adjusted p-values are given as p-values. Numbers of nodes analysed in each of the benign and malignant groups are shown on the x-axis.

	n+*	n-*	Sensitivity (%)	Specificity (%)	Optimal cutpoint	AUC (95% CI)	PPV	NPV	PLR	NLR	Youden's J Index	Accuracy	
FDG-PET/CT	SUVmax	24	36	83%	67%	3.6	0.84 (0.74-0.94)	63%	85%	2.50	0.25	0.50	73%
	NTR	23	35	65%	94%	0.47	0.80 (0.67-0.92)	88%	80%	11.41	0.37	0.60	83%
	STAR	22	30	68%	90%	9.27	0.86 (0.76-0.96)	83%	79%	6.82	0.35	0.58	81%
FEC-PET/CT	SUVmax	9	22	78%	77%	2.7	0.76 (0.54-0.98)	58%	89%	3.42	0.29	0.55	77%
	NTR	9	19	67%	74%	0.38	0.68 (0.45-0.91)	54%	82%	2.53	0.45	0.40	71%
	STAR	8	14	88%	64%	3.4	0.79 (0.61-0.98)	58%	90%	2.45	0.19	0.52	73%
DW-MRI	ADCmean (mm²/s)	28	78	46%	86%	0.85	0.63 (0.51-0.74)	90%	36%	3.23	0.63	0.32	57%
	NTR	23	56	61%	59%	1.28	0.55 (0.41-0.69)	38%	79%	1.48	0.66	0.20	59%

Table 2 - Diagnostic performance of FDG-PET/CT, FEC-PET/CT and DW-MRI from ROC analysis in endometrial cancer. * = number of positive (n+) and negative (n-) assessed lymph nodes. ADCmean is measured in x 10⁻³ mm²/s. NTR, nodal-to-tumour ratio; STAR, SUVmax-to-ADCmean ratio; AUC, area under the curve; PPV, positive predictive value; NPV, negative predictive value; PLR, positive likelihood ratio; NLR, negative likelihood ratio.

accuracy and Youden's J index. NTR performed poorly with a relatively low sensitivity, specificity, and AUC. STAR exhibited a high sensitivity and low specificity. Linear regression analysis showed significant correlation between SUVmax and SA diameter ($p < 0.01$). However, no significant correlation was observed between SUVmax and histology with or without adjustment for SA diameter.

In cervical cancer, very low numbers of analysed nodes limited reliability of results. The results from the ROC analysis are summarised in Table S1 and corresponding ROC curves plotted in Figure S4B.

Diagnostic performance of DW-MRI

The ROC analysis results from DW-MRI are summarised in Table 2 and ROC curves plotted in Figure 5C. DW-MRI, using both ADCmean and NTR, performed poorly and exhibited low accuracy and Youden's J index. ADCmean was not significantly correlated with SA diameter when analysed with linear regression. No significant correlation between ADCmean and histology was found with logistic regression analysis.

The results from the cervical cancer cohort are summarised in Table S1 and corresponding ROC curves in Figure S4C. In cervical cancer, use of an ADCmean cut-off of $0.83 \times 10^{-3} \text{ mm}^2/\text{s}$ produced high specificity and low sensitivity yielding a very high PPV. Interestingly, using a NTR cut-off of 1.15 did not misclassify any benign nodes, producing a specificity of 100%. Sensitivity was also low with NTR.

Interobserver agreement

Intraclass correlation coefficients (ICC) were calculated for each imaging modality. FDG-PET/CT and FEC-PET/CT had excellent interobserver agreement at 0.999 (95% CI 0.998-0.999) and 0.983 (95% CI 0.963-0.992) respectively. DW-MRI had poor interobserver agreement at 0.549 (95% CI 0.345-0.702). Interestingly, the interobserver agreement between ADCmean measurements of the primary tumour on DW-MRI was higher with an ICC of 0.797 (95% CI 0.546-0.916).

Readers were more likely to take a measurement in metastatic regions. For all imaging modalities, the frequency of both readers providing a measurement in metastatic regions was higher than the frequency of both readers providing a measurement in benign regions (FDG-PET/CT: 90% vs 67%; FEC-PET/CT: 83% vs 57%; DW-MRI: 70% vs 31%).

Discussion

The main finding of the present study is that SUVmax, measured on FDG-PET/CT, is significantly higher in metastatic lymph nodes of the pelvis compared to benign nodes in endometrial cancer. There is no difference between ADCmean in metastatic and benign lymph nodes in endometrial or cervical cancer. Furthermore, utilising nodal-to-tumour ratio and SUVmax-to-ADCmean ratio on FDG-PET/CT improves the detection of metastatic nodes and may prove useful in the diagnosis of lymph node metastasis.

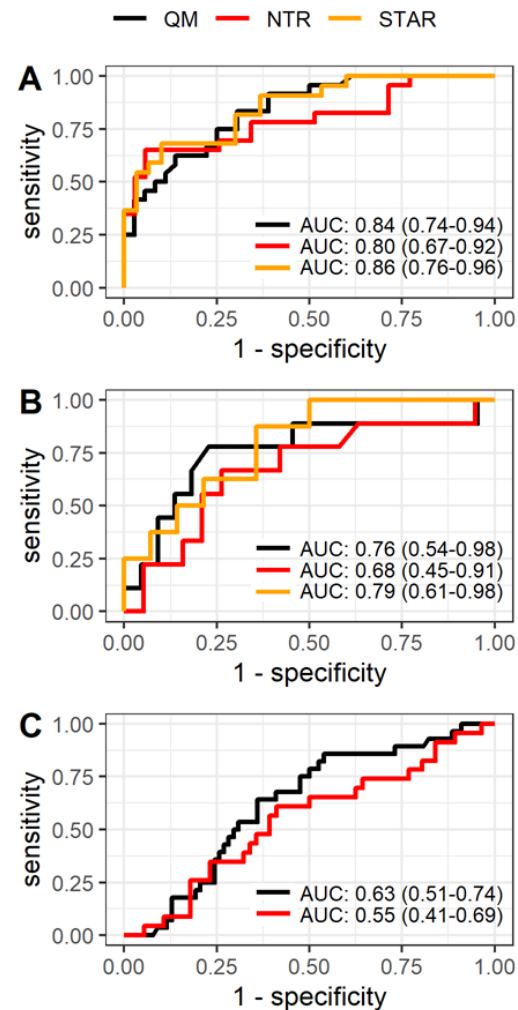


Figure 5 - ROC curves from (A) FDG-PET/CT, (B) FEC-PET/CT, and (C) DW-MRI in endometrial cancer. Plotted curves are (QM) raw quantitative measures (SUVmax for FDG-PET/CT and FEC-PET/CT and ADCmean for DW-MRI), (NTR) nodal-to-tumour ratio, and (STAR) SUVmax-to-ADCmean ratio. Corresponding area under the curve (AUC) for each measure (coloured bars) is plotted alongside 95% confidence intervals.

Using a SUVmax cut-off of 3.6 from ROC analysis on FDG-PET/CT in endometrial cancer yielded a sensitivity of 83% and specificity of 67%. Visual assessment of FDG-PET/CT is commonly regarded to be a useful diagnostic tool in ruling out lymph node metastases as previous studies have found high specificity and low sensitivity of detection (11, 23). Kim et al. retrospectively investigated the diagnostic performance of SUVmax on FDG-PET/CT in an endometrial cancer cohort and, using a cut-off of 2.8, recorded a sensitivity of 69% and specificity of 92% (11). Using SUVmax, this study recorded a higher sensitivity but a lower specificity.

The association between ADCmean/ADCmin and SUVmax of primary endometrial and cervical cancer tumours has been investigated in a few small studies. No studies to-date have investigated STAR or NTR of lymph nodes in the detection of metastasis in endometrial or cervical cancer. It has been shown that SUVmax and ADCmean of the primary tumour is inversely correlated in endometrial cancer, with significantly elevated STAR in high grade cancers, and is strongly correlated with the biomarkers of

proliferation in cervical cancer (24, 25). A recent study reported the diagnostic performance of STAR of the primary tumour exceeded both SUVmax (from FDG-PET/CT) and ADCmean in the prediction of lymph node metastases in cervical cancer with an AUC of 0.81 (95% CI 0.57-1.0) (26). In this study, nodal-to-tumour ratio (NTR) and nodal SUVmax-to-ADCmean ratio (STAR) altered the diagnostic performance of FDG-PET/CT. STAR outperformed SUVmax in cervical cancer in this study, with a sensitivity, specificity, and AUC of 80%, 83% and 0.87 (95% CI 0.64-1.0) respectively. Using an SUVmax NTR cut-off of 0.47 produced a very high specificity (94%), comparable to sensitivity reported from large prospective trials using visual assessment of FDG-PET/CT scans, and may therefore be useful in ruling out lymph node metastases (27). Interestingly, another study has found the inverse to be true in non-small cell lung cancer. Investigating the diagnostic performance of NTR in the detection of metastatic mediastinal lymph nodes, an NTR cut-off of 0.56 produced a sensitivity of 94% and specificity of 72% (28). Additionally, NTR and STAR produced the highest PPV at 88% and 83% respectively suggesting lymph nodes exhibiting an NTR or STAR over 0.47 and 9.3 respectively are often metastatic. In comparison to visual assessment by readers reported previously in the literature, this PPV is much higher. Antonsen et al. reported a PPV of just 59% from a large multicentre prospective trial and another retrospective trial reported a PPV of 76% (11, 27). Therefore, use of NTR and STAR may improve the reliability of metastatic lymph node diagnosis.

Lower numbers of cervical cancer patients limited the number of nodes available for analysis and therefore the reliability of the results. Additionally, 88% of cervical cancer patients were diagnosed with stage IB1 cancer limiting the applicability of these results to higher grade cancers. In cervical cancer, a retrospective study found a normalised SUVmax, an unvalidated method of normalising SUVmax to the pelvic blood pool, cut-off of greater than 1 produced a sensitivity of 53% and sensitivity of 75% (29). The results here suggest an SUVmax cut-off of 2.6 on FDG-PET/CT can produce a higher sensitivity but lower specificity. STAR cut-off of 4.86 produced relatively high sensitivity, specificity, PPV and NPV (80%, 83%, 80%, 83% respectively) but low number of nodes analysed limit the applicability of these results.

Use of SUVmax on FEC-PET/CT for the diagnosis of malignant lymph nodes has not yet been studied in endometrial or cervical cancer. FEC-PET/CT was not mandatory for study recruitment and thus the results of this analysis are limited by low numbers of analysed nodes, especially in cervical cancer. There was a trend of mean SUVmax of metastatic nodes in the pelvis being higher in both endometrial and cervical cancer although this difference did not reach significance in either cohort. Overall, use of SUVmax, NTR and STAR on FEC-PET/CT was not as accurate as FDG-PET/CT. In general, the tumour and nodal SUVmax levels are lower with FEC-PET/CT than FDG-PET/CT. SUVmax and STAR produced a high NPV (89% and 90% respectively) narrowly outperforming other quantitative measures on FDG-PET/CT and DW-MRI suggesting these measures may be useful in ruling out lymph node metastases. That being said, visual assessment

of FDG-PET/CT scans regularly outperforms use of quantitative measures from FEC-PET/CT in terms of NPV and specificity and, combined with the reduced use and availability of FEC-PET/CT, makes utilising these quantitative measures from FEC-PET/CT less practical (11, 27).

Multiple prospective and retrospective studies from the past two decades have shown various measurements of ADC (ADCmin, ADCmean and ADCmax) to be significantly lower in metastatic lymph nodes in both cervical and endometrial cancer (17, 18, 30-33). Nakai et al. found no significant difference in ADC between metastatic and benign lymph nodes although this study was conducted on a small sample size of patients with endometrial or cervical cancer and measurements of ADC taken by a single reader (15). Additionally, another large prospective study in multiple cancer types (including non-gynaecological cancers) found no significant difference in ADCmean between metastatic and benign nodes (16).

No significant difference in ADCmean was observed between metastatic and benign lymph nodes in the present study. In general, this contradicts findings from recent literature. Interobserver agreement of ADCmean measurements of the primary tumour was low with an intraclass correlation coefficient (ICC) of 0.797. Interestingly, where reported, previous studies had excellent interobserver agreement of ADCmean measurements in contrast to this study which had a poor ICC of 0.549 (18, 32, 33). Previous literature reporting excellent interobserver agreement of ADCmean measurements of lymph nodes selected lymph nodes on T2-weighted imaging and measured lymph nodes on a node-by-node basis compared with selecting the most restricted node by region as in this study. This may provide an explanation as to why the results presented here are not in concordance with previously reported findings and highlights an area for improvement in the protocol.

Previous studies have reported strong diagnostic performance of DW-MRI when using an ADCmean cut-off. Sensitivities range from 79% to 100% and specificities from 72% to 97% with better diagnostic performance obtained in endometrial cancer than cervical cancer (17, 18, 30-33). In this study, utilising ADCmean and NTR cut-offs of $0.852 \times 10^{-3} \text{ mm}^2/\text{s}$ and 1.28 respectively, DW-MRI performed poorly in endometrial cancer. Although sensitivity for ADCmean was low, the PPV was 90% suggesting a nodal ADCmean greater than this cut-off can be reliably used to diagnose malignant lymph nodes. Furthermore, a previous retrospective study investigating the diagnostic performance of ADCmean on DW-MRI reported a lower PPV in endometrial cancer (18). Additionally, this high PPV represents an improvement on previously reported PPVs from visual assessment of lymph nodes by expert readers, suggesting use of ADCmean on DW-MRI may be a beneficial adjunct in the diagnosis of metastatic lymph nodes in endometrial cancer (11, 33).

In cervical cancer, an ADCmean cut-off of $0.830 \times 10^{-3} \text{ mm}^2/\text{s}$ gave a specificity of 89%. Interestingly, employing an NTR cut-off of 1.15 meant all benign nodes were correctly classified, producing a specificity of 100%,

suggesting these measures may be useful in ruling out lymph node metastases in cervical cancer. Additionally, both ADCmean and NTR had high PPV (95% and 100% respectively). Overall, relatively low numbers of nodes in cervical cancer limit the reliability of these results and further studies would be necessary to validate these high specificities and PPVs.

Overall, the evidence presented here suggests quantitative measurements from FDG-PET/CT, FEC-PET/CT and DW-MRI cannot be reliably used alone to diagnose lymph node metastases in endometrial and cervical cancer.

Quantitative measures, including NTR and STAR, may represent a useful tool in specific scenarios when used in adjunction with visual assessment by an expert reader to improve the detection and reliability of diagnosis of lymph node metastases.

This study has a number of limitations. Firstly, lymph nodes were dissected en bloc from the left pelvis, right pelvis, and para-aortic region. From histology, a single malignant node in a region classified that region as malignant. Therefore, it is difficult to determine if a node classified as malignant by histology was the node being quantified on the scans. Secondly, readers often quantified multiple nodes during examination of the scans; it was possible for each reader to quantify different nodes. To remove bias and standardise the protocol, the most intense measurement was selected from each region. It was therefore possible for different nodes to be selected for analysis. This may somewhat account the low interobserver agreement observed with the measurement of ADCmean on DW-MRI. Thirdly, some patients only received quantitative measurements from a single reader. Again, this raises an issue of reliability of results particularly for DW-MRI where poor interobserver agreement was observed. A lower percentage of benign regions being measured by two readers suggests that readers were biased to take measurements from malignant regions. Improvement of the protocol for obtaining quantitative measurements would help mitigate these issues in future studies. Finally, the limited spatial resolution of FDG-PET/CT and FEC-PET/CT may have led to a partial volume effect in small lymph nodes and so SUVmax in these smaller nodes may be underestimated.

To determine if multiple measures combine to improve diagnostic performance in the detection of metastatic lymph nodes further work will be carried out. The aim of this future work will be to combine measures described in the present study to maximise sensitivity and specificity through synthesis of a nomogram.

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Abbreviations:

ADCmax, maximum apparent diffusion coefficient; ADCmean, mean apparent diffusion coefficient; ADCmin, minimum apparent diffusion coefficient; AUC, area under the curve; CT, computed tomography; DOR, diagnostic odds ratio; DW-MRI, diffusion weighted magnetic resonance imaging; FDG, ¹⁸F-fluoro-2-deoxy-D-glucose; FDR, false discovery rate; FEC, ¹⁸F-fluoroethyl-choline; FIGO, The International Federation of Gynaecology and Obstetrics; ICC, intraclass correlation coefficient; MRI, magnetic resonance imaging; NLR, negative likelihood ratio; NPV, negative predictive value; NTR, nodal-to-tumour ratio; PET, positron emission tomography; PLR, positive likelihood ratio; PPV, positive predictive value; QM, raw quantitative measure; ROC, receiver operating characteristic; ROI, region of interest; STAR, SUVmax-to-ADCmean ratio; SUVmax, maximum standardised uptake value.

Supplementary Code:

https://github.com/cancer-sci/proj_1_finalcode

Supplementary Material

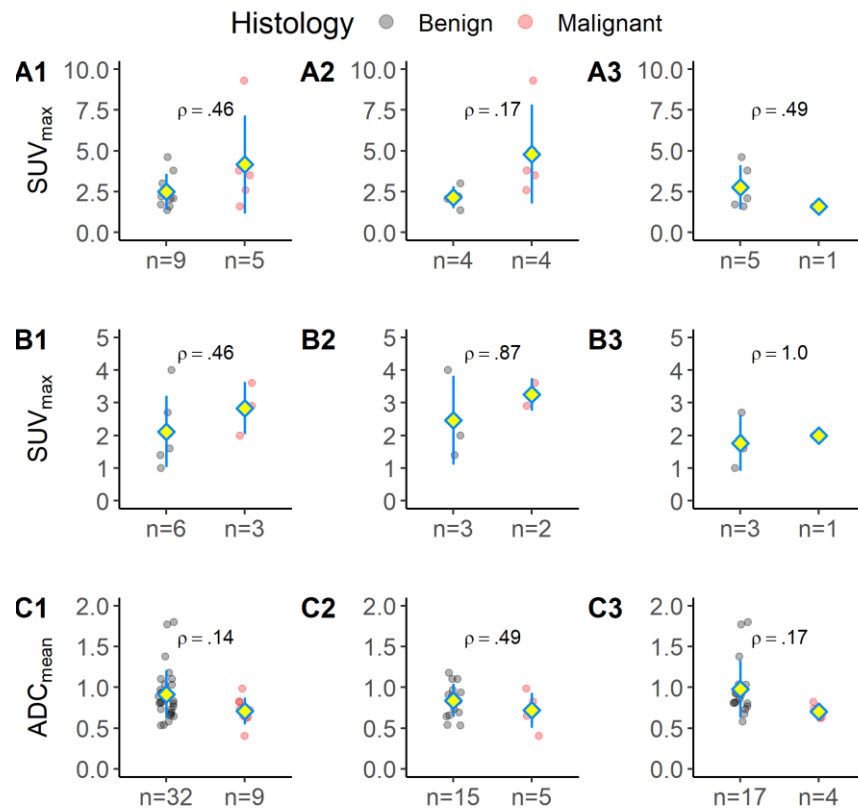


Figure S1 - Differences in SUVmax on (A) FDG-PET/CT and (B) FEC-PET/CT and in ADCmean (mm²/s) on (C) DW-MRI in benign and malignant lymph nodes in patients with cervical cancer. Column 1 represents pelvic nodes pooled together, column 2 left pelvic nodes, and column 3 represents right pelvic lymph nodes. Yellow points represent the group mean and the blue lines represent ± 1 standard deviation. Significant difference between group means, calculated by Mann-Whitney U-test, is represented with a bracket and asterisk (*; $p < 0.05$). FDR adjusted p-values are given as p-values. Numbers of nodes analysed in each of the benign and malignant groups are shown on the x-axis.

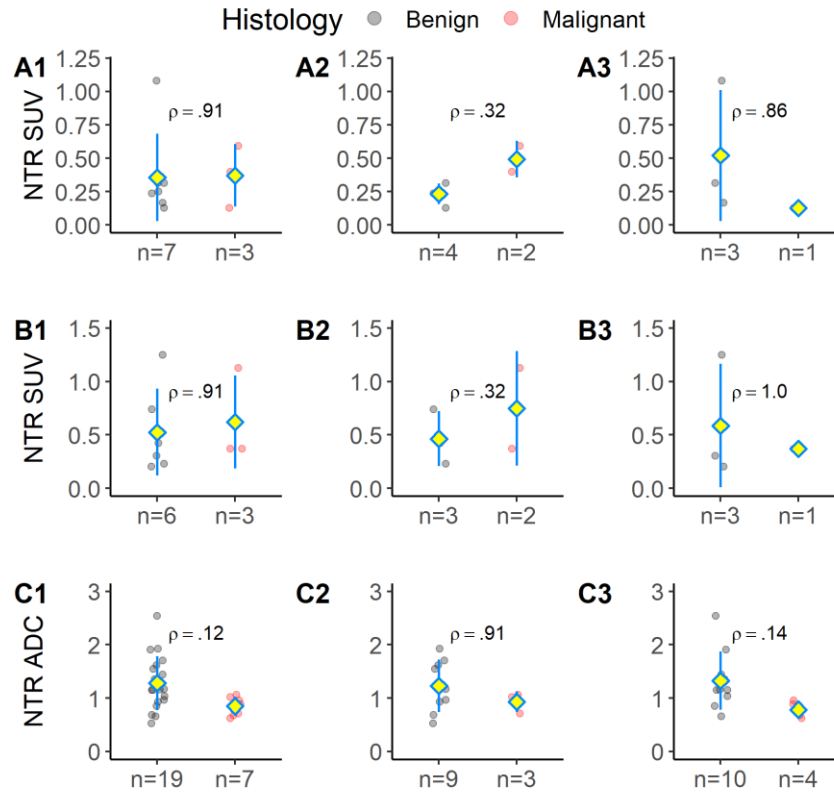


Figure S2 - Nodal-to-tumour ratio (NTR) of SUVmax on (A) FDG-PET/CT and (B) FEC-PET/CT and of ADCmean (mm²/s) on (C) DW-MRI in benign and malignant lymph nodes in patients with cervical cancer. Column 1 represents pelvic nodes pooled together, column 2 left pelvic nodes, and column 3 represents right pelvic lymph nodes. Yellow points represent the group mean and the blue lines represent ± 1 standard deviation. Significant difference between group means, calculated by Mann-Whitney U-test, is represented with a bracket and asterisk (*; $p < 0.05$). FDR adjusted p-values are given as p-values. Numbers of nodes analysed in each of the benign and malignant groups are shown on the x-axis.

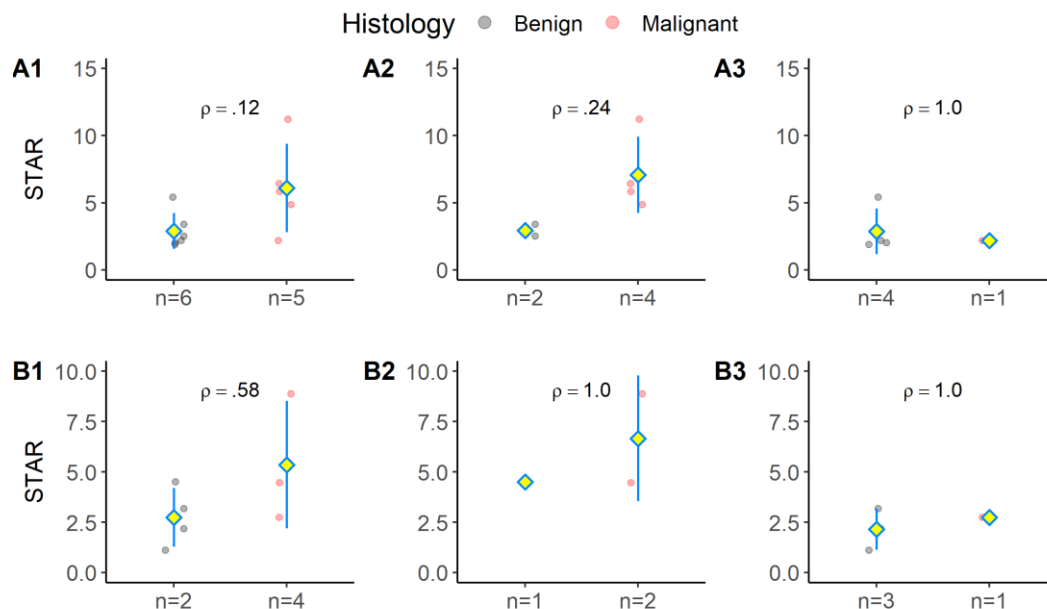


Figure S3 - SUVmax-to-ADC ratio (STAR) from combined (A) FDG-PET/CT and DW-MRI and (B) FEC-PET/CT and DW-MRI in benign and malignant lymph nodes in patients with cervical cancer. Column 1 represents pelvic nodes pooled together, column 2 left pelvic nodes, and column 3 represents right pelvic lymph nodes. Yellow points represent the group mean and the blue lines represent ± 1 standard deviation. Significant difference between group means, calculated by Mann-Whitney U-test, is represented with a bracket and asterisk (*; $p < 0.05$). FDR adjusted p-values are given as p-values. Numbers of nodes analysed in each of the benign and malignant groups are shown on the x-axis.

		n+*	n-*	Sensitivity (%)	Specificity (%)	Optimal cutpoint	AUC (95% CI)	PPV	NPV	PLR	NLR	Youden's J index	Accuracy
FDG-PET/CT	SUVmax	5	9	80%	67%	2.6	0.69 (0.35-1.0)	57%	86%	2.40	0.30	0.47	71%
	NTR	3	7	67%	86%	0.40	0.57 (0.79-0.97)	67%	86%	4.67	0.39	0.52	80%
	STAR	5	6	80%	83%	4.86	0.87 (0.64-1.0)	80%	83%	4.80	0.24	0.63	82%
FEC-PET/CT	SUVmax	3	6	67%	67%	2.45	0.75 (0.4-1.0)	50%	80%	2.00	0.50	0.33	67%
	NTR	3	6	100%	50%	0.37	0.61 (0.2-1.0)	50%	100%	2.00	0.00	0.50	67%
	STAR	3	4	100%	50%	2.73	0.75 (0.33-1.0)	60%	100%	2.00	0.00	0.50	71%
DW-MRI	ADCmean	9	32	56%	89%	0.830	0.73 (0.55-0.91)	95%	36%	5.06	0.49	0.45	63%
	NTR	7	19	63%	100%	1.15	0.79 (0.62-0.96)	100%	50%	N/A	0.37	0.63	73%

Table S1 - Diagnostic performance of FDG-PET/CT, FEC-PET/CT and DW-MRI from ROC analysis in cervical cancer. * = number of positive (n+) and negative (n-) assessed lymph nodes. ADCmean is measured in $\times 10^{-3} \text{ mm}^2/\text{s}$. NTR, nodal-to-tumour ratio; STAR, SUVmax-to-ADCmean ratio; AUC, area under the curve; PPV, positive predictive value; NPV, negative predictive value; PLR, positive likelihood ratio; NLR, negative likelihood ratio.

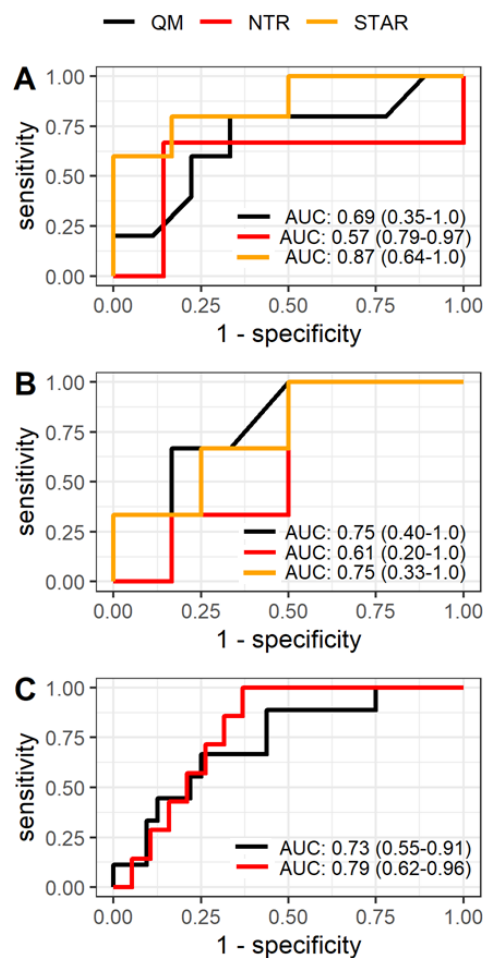


Figure S4 -ROC curves from (A) FDG-PET/CT, (B) FEC-PET/CT, and (C) DW-MRI in cervical cancer. Plotted curves are (QM) raw quantitative measures (SUVmax for FDG-PET/CT and FEC-PET/CT and ADCmean for DW-MRI), (NTR) nodal-to-tumour ratio, and (STAR) SUVmax-to-ADCmean ratio. Corresponding area under the curve (AUC) for each measure (coloured bars) is plotted alongside 95% confidence intervals.